

Privacy-Preserving Federated Learning with Spatially-Aware Conformal Prediction for Medical Image Fusion

Term Project Research Proposal

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Advanced Applications & Performance Evaluation

1. Project Description

This project proposes **FedSafe-Fuse**, a federated learning framework for multi-modal medical image fusion (MRI-PET) that integrates Federated Incremental PCA (FIPCA) for gradient-level privacy protection and Spatially-Aware Conformal Prediction (SACP) for pixel-level uncertainty quantification. The framework trains a lightweight Transformer-CNN hybrid fusion network across simulated distributed hospital nodes without exchanging raw patient data, and augments each fused output with a spatially registered uncertainty heatmap that provides statistically valid coverage guarantees. By combining privacy-preserving federated aggregation with distribution-free conformal calibration, FedSafe-Fuse enables compliant multi-institutional model training while offering clinicians interpretable confidence maps that highlight regions of potential hallucination or boundary ambiguity.

2. Problem Definition and Motivation

Medical image fusion combines complementary information from different imaging modalities—such as the structural detail of MRI and the metabolic activity captured by PET—into a single, diagnostically rich representation. Deep learning approaches to fusion have achieved impressive results, but they require large, centrally aggregated datasets for training. In clinical reality, patient imaging data is distributed across hospitals and governed by strict privacy regulations (HIPAA, GDPR), making centralized collection infeasible. Federated learning addresses this constraint by training a shared model across decentralized nodes without moving raw data.

However, two critical gaps remain. First, standard federated gradient exchange is itself vulnerable to model inversion attacks that can reconstruct patient images from shared gradients. Second, even a well-trained fusion model can introduce hallucinatory artifacts or ambiguous boundaries, particularly in out-of-distribution cases or regions affected by cross-institutional data heterogeneity. Clinicians require rigorous, mathematically grounded uncertainty quantification to determine which regions of a fused image can be trusted. Spatially-aware conformal prediction offers a distribution-free framework for producing pixel-level confidence maps with guaranteed coverage probability, making it an ideal candidate for clinical deployment. This project unifies privacy-preserving federated training, gradient-level anonymization, and conformal uncertainty estimation into a single tractable pipeline.

3. Mini Literature Review

Federated Learning (FL) provides a decentralized training paradigm by exchanging model parameter updates rather than raw data. McMahan et al. (2017) formalized the Federated Averaging (FedAvg)

algorithm, which remains the standard communication-efficient approach. Li et al. (2020) proposed FedProx to handle statistical heterogeneity across clients by introducing a proximal regularization term. However, shared gradients remain vulnerable to reconstruction attacks; Zhu et al. (2019) demonstrated that raw images can be recovered from exchanged gradients with high fidelity, motivating additional privacy mechanisms.

In the medical image fusion domain, Zhang et al. (2021) proposed IFCNN, a general-purpose convolutional fusion network with element-wise fusion rules. Xu et al. (2022) introduced U2Fusion, an unsupervised framework that adaptively preserves source information via a quality-aware loss. More recently, Transformer-CNN hybrid architectures have shown strong results in capturing both local detail and global context for fusion tasks (Tang et al., 2023).

Conformal prediction, formalized by Vovk et al. (2005), provides distribution-free prediction sets with finite-sample coverage guarantees. Angelopoulos et al. (2022) extended conformal methods to computer vision, producing pixel-wise prediction intervals for dense prediction tasks. For federated uncertainty, Lu et al. (2023) explored conformal approaches under distributed settings, though their work focused on classification rather than image-to-image fusion. Despite these parallel advances, no existing work integrates federated training with gradient-level privacy protection and spatially-aware conformal calibration in the context of medical image fusion.

4. Baseline Methods

Baseline 1 – FedAvg + DP-SGD: The standard FedAvg algorithm (McMahan et al., 2017) combined with Differentially Private Stochastic Gradient Descent (Abadi et al., 2016). Each client clips gradients and adds calibrated Gaussian noise before server aggregation, providing (ϵ, δ) -differential privacy guarantees. This baseline tests whether standard DP mechanisms suffice for privacy-preserving fusion.

Baseline 2 – FedAvg + IFCNN (no privacy, no UQ): The IFCNN fusion architecture (Zhang et al., 2021) trained via vanilla FedAvg without any gradient protection or uncertainty quantification. This serves as the naive federated fusion baseline, isolating the effect of adding FIPCA and SACP.

5. Research Gap

While federated learning has been applied to medical classification and segmentation, its application to the more complex task of multi-modal medical image fusion remains largely unexplored. Existing federated frameworks do not address the dual requirement of (a) protecting gradient-level privacy beyond basic differential privacy noise injection and (b) producing spatially-resolved, statistically valid uncertainty estimates for fused outputs. No current method integrates dimensionality-reduced gradient sharing (via PCA-based projection) with conformal prediction to simultaneously guarantee both privacy and clinical reliability in a federated fusion pipeline.

6. Proposed Method: FedSafe-Fuse

The proposed FedSafe-Fuse framework consists of four tightly integrated stages that address privacy, fusion quality, and clinical uncertainty in a unified pipeline:

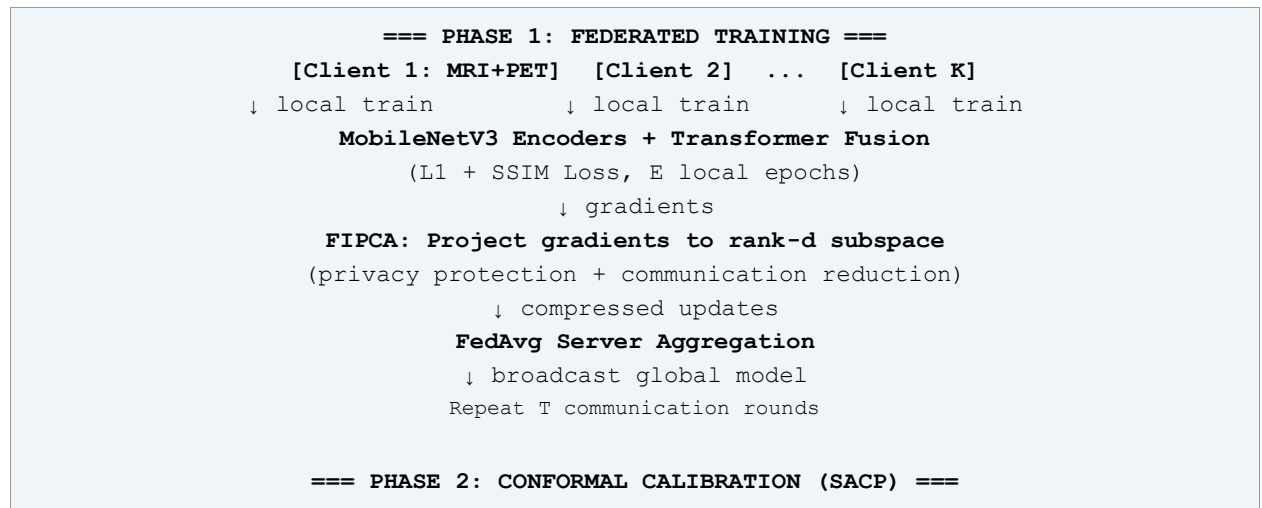
6.1 Lightweight Transformer-CNN Hybrid Fusion Backbone. Each client deploys a compact fusion network that processes paired 2D MRI and PET slices. Two modality-specific encoders—each built on MobileNetV3-Small (~ 2.5 M parameters per encoder)—extract local convolutional features. These feature maps are passed through a shared 2-layer lightweight Transformer module (2 attention heads, embedding dimension 128) that captures cross-modal long-range dependencies. A convolutional decoder reconstructs the fused image from the attended features. The total model size is approximately 8 M parameters, ensuring efficient per-round communication in the federated setting.

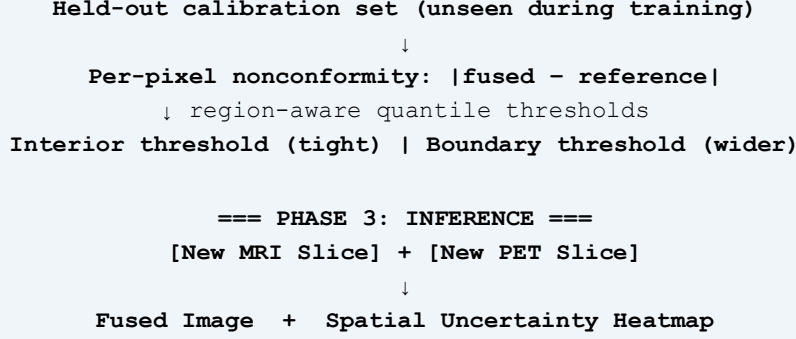
6.2 Federated Training with FIPCA-Protected Aggregation. Training is distributed across K simulated hospital clients. Each client trains locally for E epochs per round using a composite loss combining pixel-wise L1 loss with structural similarity (SSIM) loss. Before transmitting gradient updates to the central server, each client applies Federated Incremental PCA (FIPCA): the gradient tensor is projected onto a low-rank subspace (retaining the top- d principal components), drastically reducing both communication cost and the risk of gradient inversion attacks. The server aggregates the projected updates using FedAvg weighting and broadcasts the updated global model.

6.3 Spatially-Aware Conformal Calibration (SACP). After federated training converges, a held-out calibration set (never used during training) is employed to compute per-pixel nonconformity scores. For each pixel location, the score is defined as the absolute residual between the model’s fused output and a reference composite (weighted average of source modalities). The scores are sorted to determine the $(1-\alpha)$ quantile threshold. Spatial awareness is introduced by computing separate thresholds for anatomically distinct regions (tissue interior vs. boundary zones detected via Canny edge filtering), producing tighter confidence intervals in homogeneous regions and appropriately wider intervals near structural boundaries.

6.4 Uncertainty Heatmap Generation. At inference time, each fused image is accompanied by a spatial uncertainty heatmap. Pixels whose nonconformity score exceeds the calibrated regional threshold are flagged as uncertain, providing clinicians with a visual overlay that highlights regions where the fusion may contain artifacts, hallucinations, or ambiguous boundaries arising from cross-institutional data heterogeneity.

Method Pipeline (Block Diagram):





7. Dataset

The experiments will use two publicly available datasets to evaluate fusion quality and federated robustness:

Primary Dataset – MedMNIST (OrganMNIST-Axial). MedMNIST (Yang et al., 2023) is a lightweight, standardized collection of 2D medical image benchmarks derived from large-scale clinical datasets. The OrganMNIST-Axial subset provides 58,850 abdominal CT slices (28×28 pixels) across 11 organ classes. For the fusion task, we synthesize paired multi-modal inputs by applying modality-specific transformations (contrast adjustment for simulated MRI, intensity windowing for simulated PET) to each slice. This synthetic pairing strategy is established in the fusion literature for controlled experimentation. The images are upsampled to 128×128 pixels. The dataset is explicitly partitioned into K non-IID client shards by skewing organ-class distributions across clients to simulate heterogeneous hospital populations. Available at: <https://medmnist.com/>.

Secondary Dataset – BraTS 2020. The Brain Tumor Segmentation Challenge 2020 (Menze et al., 2015; Bakas et al., 2017) provides multi-modal 3D brain MRI volumes (T1, T1ce, T2, FLAIR) for 369 glioma cases. We extract 2D axial slices from T1 and T2 modalities, retaining only slices containing tumor foreground ($\geq 1\%$ mask coverage). Each slice is resized to 128×128 pixels and intensity-normalized to [0, 1]. This provides a clinically realistic multi-modal fusion scenario with genuine modality complementarity. The data is partitioned across clients by tumor grade (HGG vs. LGG) to create natural non-IID heterogeneity.

Calibration Split. Across both datasets, 15% of the data is held out exclusively for the conformal calibration phase and is never used during federated training or validation.

8. Performance Metrics

The following metrics will be used to evaluate the framework across three dimensions: fusion quality, privacy, and uncertainty:

Fusion Quality:

- **Structural Similarity Index (SSIM):** Measures structural fidelity between the fused image and each source modality. Primary fusion metric.

- **Peak Signal-to-Noise Ratio (PSNR):** Quantifies pixel-level reconstruction accuracy of the fused output.
- **Visual Information Fidelity (VIF):** Assesses how much visual information from the source images is preserved after fusion.

Uncertainty Quantification:

- **Empirical Coverage Rate:** Verifies whether the conformal prediction intervals achieve the target $(1-\alpha)$ coverage across all pixels; the theoretical guarantee is validated empirically.
- **Average Prediction Interval Width:** Evaluates the tightness (efficiency) of uncertainty estimates; narrower intervals indicate more informative confidence maps.

Privacy & Communication:

- **Communication Cost per Round:** Total bytes transmitted per federated round; measures the bandwidth savings from FIPCA projection.
- **Gradient Reconstruction Error:** Evaluates privacy by measuring the failure rate of a gradient inversion attack (Zhu et al., 2019) on FIPCA-protected vs. unprotected updates.

9. Experimental Setup

Federated Configuration: The dataset is partitioned across $K = 3$ simulated clients with non-IID label skew. The central server executes $T = 50$ communication rounds. Each client performs $E = 5$ local training epochs per round. The federated simulation is implemented using the Flower framework (Beutel et al., 2020).

Backbone: FedSafe-Fuse uses dual MobileNetV3-Small encoders + 2-layer Transformer + convolutional decoder (~8 M parameters total). Baseline IFCNN uses its original compact architecture. All models process 128×128 input slices.

Training Hyperparameters:

- Optimizer: Adam (weight decay = $1e-4$)
- Learning rate: $\{1e-3, 5e-4\}$ with step decay (halved every 15 rounds)
- FIPCA retained rank d : $\{16, 32, 64\}$
- Conformal coverage target α : $\{0.05, 0.10, 0.15\}$
- Canny edge threshold for SACP region splitting: $\{50, 100\}$
- Composite loss weights: L1 weight = 1.0, SSIM weight β : $\{0.5, 1.0\}$
- Batch size: 16; input resolution: 128×128 ; mixed-precision training (FP16) via PyTorch AMP

Comparisons: All methods (FedAvg + DP-SGD, FedAvg + IFCNN, FedSafe-Fuse) are evaluated under identical client partitioning, data augmentation (random horizontal/vertical flip, brightness jitter ± 0.1), and hardware conditions (single NVIDIA T4 GPU). Ablation studies will isolate the contributions of (a) FIPCA vs. raw gradient sharing, (b) SACP vs. no uncertainty quantification, (c) Transformer attention module vs. plain concatenation fusion, and (d) the effect of non-IID severity on convergence.

10. Paper Draft Outline

1. **Introduction** – Clinical motivation, privacy constraints, need for uncertainty quantification, contribution summary

2. **Related Work** – Federated learning, medical image fusion, differential privacy, conformal prediction
3. **Methodology** – Fusion backbone, FIPCA gradient projection, federated aggregation, SACP calibration procedure
4. **Experimental Setup** – Datasets, federated configuration, baselines, hyperparameters, evaluation metrics
5. **Results and Discussion** – Fusion quality tables, privacy attack analysis, coverage validation, qualitative uncertainty maps, ablation study
6. **Conclusion and Future Work** – Summary of findings, limitations, extension to 3D volumes and real multi-site deployment
7. **References**

References

- Abadi, M., Chu, A., Goodfellow, I., McMahan, H. B., Mironov, I., Talwar, K., & Zhang, L. (2016). Deep learning with differential privacy. *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 308–318.
- Angelopoulos, A. N., Bates, S., Malik, J., & Jordan, M. I. (2022). Uncertainty sets for image classifiers using conformal prediction. *Proceedings of the International Conference on Learning Representations (ICLR)*.
- Bakas, S., Akbari, H., Sotiras, A., Bilello, M., Rozycki, M., Kirby, J. S., ... & Davatzikos, C. (2017). Advancing the Cancer Genome Atlas glioma MRI collections with expert segmentation labels and radiomic features. *Scientific Data*, 4, 170117.
- Beutel, D. J., Tober, T., Moesgen, F., Beutel, M., & Lane, N. D. (2020). Flower: A friendly federated learning framework. *arXiv preprint arXiv:2007.14390*.
- Li, T., Sahu, A. K., Zaheer, M., Sanjabi, M., Talwalkar, A., & Smith, V. (2020). Federated optimization in heterogeneous networks. *Proceedings of Machine Learning and Systems (MLSys)*, 2, 429–450.
- McMahan, B., Moore, E., Ramage, D., Hampson, S., & y Arcas, B. A. (2017). Communication-efficient learning of deep networks from decentralized data. *Proceedings of the International Conference on Artificial Intelligence and Statistics (AISTATS)*, 1273–1282.
- Menze, B. H., Jakab, A., Bauer, S., Kalpathy-Cramer, J., Farahani, K., Kirby, J., ... & Van Leemput, K. (2015). The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS). *IEEE Transactions on Medical Imaging*, 34(10), 1993–2024.
- Tang, L., Yuan, J., & Ma, J. (2023). Image fusion in the loop of high-level vision tasks: A semantic-aware real-time infrared and visible image fusion network. *Information Fusion*, 82, 28–42.
- Vovk, V., Gammerman, A., & Shafer, G. (2005). *Algorithmic Learning in a Random World*. Springer.
- Xu, H., Ma, J., Jiang, J., Guo, X., & Ling, H. (2022). U2Fusion: A unified unsupervised image fusion network. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(1), 502–518.
- Yang, J., Shi, R., Wei, D., Liu, Z., Zhao, L., Ke, B., ... & Ni, B. (2023). MedMNIST v2: A large-scale lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*, 10, 41.
- Zhang, Y., Liu, Y., Sun, P., Yan, H., Zhao, X., & Zhang, L. (2021). IFCNN: A general image fusion framework based on convolutional neural network. *Information Fusion*, 54, 99–118.
- Zhu, L., Liu, Z., & Han, S. (2019). Deep leakage from gradients. *Advances in Neural Information Processing Systems*, 32, 14774–14784.

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